

Matched controls and held-out evaluation reveal unstable Mamba attribution in YOLO26 traffic-object detection

Abstract

State-space mixers are increasingly used in object detectors, yet gains attributed to Mamba may also arise from tokenization, normalization, projection, residual gating, or added capacity in the surrounding branch. We tested mixer-specific attribution in YOLO26n using an unchanged baseline (C0), a gated width-sequence branch with packed bidirectional Mamba (C1-W), and a near-parameter-matched control replacing only Mamba with a kernel-8 Conv1d (C1-C8). Each variant was trained for 100 epochs with three paired seeds on sequence-disjoint UA-DETRAC-G2 and independently on VisDrone. The nine development-selected checkpoints per dataset were then frozen and evaluated once on held-out partitions. On development data, C1-C8 exceeded C0 by 0.0091 mean average precision from 0.50 to 0.95 intersection over union (mAP50-95) on UA-DETRAC-G2 and 0.0009 on VisDrone, whereas C1-W minus C1-C8 was -0.0081 and -0.0004. These local-branch gains did not persist on held-out data: C1-C8 minus C0 changed to -0.0090 and -0.0016. The Mamba substitution effect changed to +0.0110 on UA-DETRAC-G2 held-out data and remained -0.0005 on VisDrone test-dev. C1-C8 was also 13.6% faster than C1-W under batch-1 half-precision native-forward measurement. Thus, development improvements from a multi-component branch did not provide reproducible evidence for a Mamba-specific benefit. Matched non-Mamba controls and independent evaluation are therefore necessary for credible component attribution. Results are descriptive for three seed pairs and do not establish universal mixer superiority.

Keywords: Object detection; state-space models; Mamba; matched controls; held-out evaluation; component attribution; generalization

1. Introduction

State-space mixers are increasingly used to enlarge the contextual receptive field of real-time visual models. Mamba makes the parameters of a structured state-space model input dependent and couples this selective mechanism with an implementation that scales linearly with sequence length (Gu and Dao, 2024). Visual adaptations have introduced bidirectional token processing, multi-route two-dimensional scans, and hierarchical multi-scale designs (Zhu et al., 2024; Liu et al., 2024; Shi et al., 2024). These developments make state-space modeling attractive for object detection, where long-range context must be balanced against latency and model size.

Attributing a detector improvement specifically to Mamba is nevertheless difficult. A visual state-space branch is not only a mixer: it also determines how a feature map is converted into tokens, which axes and directions are traversed, how directional outputs are fused, and how normalization, projection, gating, and residual reintegration are implemented. Representative Mamba-based detectors additionally alter backbone or neck organization and introduce residual, channel-interaction, cross-layer-fusion, or multimodal components (Wang et al., 2025; Chen et al., 2026; Wang et al., 2024; Li and Xiao, 2024; Lei et al., 2025). Their complete systems may be effective, but a comparison with an unmodified detector estimates the effect of the complete intervention. It cannot by itself isolate the contribution of the named mixer from that of the surrounding branch.

This distinction is important when reported gains are small and model selection is performed on a finite validation partition. Optimizing architectures and checkpoints against the same finite criterion can overfit

the selection procedure itself, producing selection bias that may be comparable to the differences between learning algorithms (Cawley and Talbot, 2010). Repeated feedback from a holdout partition also makes subsequent analysis adaptive and can erode its role as independent evidence (Dwork et al., 2015). A credible mixer-level claim therefore requires a non-Mamba mixer with a matched branch interface and comparable capacity, paired runs rather than isolated best cases, and evaluation on data unavailable to architecture and checkpoint decisions.

We apply this design to the nano-scale YOLO26 detector (Jocher et al., 2026) in two traffic-object detection settings with materially different viewpoints and taxonomies. UA-DETRAC comprises fixed-camera traffic videos with annotated vehicles (Wen et al., 2020), whereas VisDrone contains drone-captured scenes and native road-user categories (Zhu et al., 2022). In each dataset, we compare the unchanged YOLO26n baseline (C0), a zero-initialized gated width-sequence branch with packed bidirectional Mamba mixing (C1-W), and a control that preserves the insertion site, pooling, tokenization, LayerNorm, projection, residual gate, initialization, and training contract while replacing only Mamba with a near-parameter-matched kernel-8 Conv1d (C1-C8). Each variant is trained with three paired seeds. Checkpoints are selected only on development validation, frozen with the architectures, metrics, and decision rules, and then evaluated in a one-time campaign on the corresponding held-out partition.

The study is therefore a controlled attribution and generalization-boundary analysis, not a proposal for a new state-of-the-art detector. It makes three contributions. First, it isolates mixer identity inside an otherwise shared gated sequence branch and distinguishes total branch, matched local-branch, and Mamba-substitution effects. Second, it separates development evidence from one-time held-out evidence on a sequence-disjoint traffic-video benchmark and an independent aerial benchmark. Third, it reports paired-seed contrasts, class-wise behavior, and scope-controlled native-forward latency, including negative and direction-reversing results. This design tests not only whether a complete Mamba branch differs from a baseline, but also whether any difference can be assigned specifically to Mamba and whether the inferred effect survives independent evaluation.

2. Related Work

2.1 Selective state-space models for visual representation

Mamba introduced input-dependent selection into structured state-space sequence modeling together with a hardware-aware recurrent algorithm (Gu and Dao, 2024). Extending this mechanism to images requires an explicit spatial ordering. Vision Mamba uses bidirectional processing of image sequences (Zhu et al., 2024), VMamba traverses two-dimensional features along four selective-scan routes (Liu et al., 2024), and Multi-Scale VMamba studies the accuracy and redundancy consequences of scan count and scale (Shi et al., 2024). These designs show that visual state-space modeling couples mixer identity with token construction, directionality, and fusion. The present study therefore treats the complete gated branch and its internal mixer as separate experimental factors.

2.2 State-space modules in object detectors

Mamba-based object detectors place state-space operations at different architectural levels. Mamba YOLO combines an ODMamba macro-architecture with ODSSBlock and residual gated components intended to retain local information (Wang et al., 2025). AKCMamba-YOLO introduces separate state-space modules for channel interaction and cross-layer semantic fusion across the backbone and neck (Chen et al., 2026). Mamba-YOLO-World replaces the neck with a multimodal MambaFusion-PAN and

introduces parallel- and serial-guided scan flows (Wang et al., 2024). DS MYOLO combines SimVSS fusion with ECACConv and ECACSP modules in driving scenes (Li and Xiao, 2024), while MambaNeXt-YOLO couples SS2D with ConvNeXt, ResGate, clue-merge, and multi-branch pyramid components (Lei et al., 2025).

These studies evaluate complete detector designs and their component combinations. In a focused audit of the five representative 2D detectors above, the reported ablations switched modules, fusion directions, component combinations, or Mamba hyperparameters. We did not identify a trained comparison that retained the complete branch shell while replacing only the Mamba mixer with a near-parameter-matched non-Mamba alternative (Supplementary Table S4). This scoped observation motivates the counterfactual evaluated here; it does not invalidate the reported systems, prove that all prior detectors lack suitable controls, or establish that Mamba is ineffective.

2.3 Matched controls and held-out evaluation

A baseline-versus-module contrast estimates the total effect of the complete intervention. It does not isolate a named internal mechanism when tokenization, normalization, projection, gating, fusion, or capacity also change. In our design, C1-W minus C0 measures the total effect of adding the gated Mamba branch, C1-C8 minus C0 measures the effect of the matched local branch, and C1-W minus C1-C8 measures the substitution effect of Mamba relative to that control. Keeping these estimands separate prevents branch-level behavior from being reported as mixer-specific evidence.

The same separation is required between selection and generalization assessment. Validation-based comparisons are vulnerable to selection bias when architectures, checkpoints, or analysis choices are repeatedly optimized against the same data (Cawley and Talbot, 2010). Reuse of holdout feedback can similarly make later choices adaptive (Dwork et al., 2015). We therefore selected each checkpoint only on development validation, froze all evaluation contracts before held-out access, reported every held-out result regardless of direction, and prohibited held-out-driven architecture selection, threshold tuning, or additional training.

2.4 Class imbalance and reporting scope

Dense detectors face a large imbalance between background and foreground examples (Lin et al., 2017), while long-tailed recognition studies show that sparsely represented foreground categories can exhibit high uncertainty and unfavorable gradient competition (Gupta et al., 2019; Tan et al., 2020). These observations motivate class-wise reporting but do not justify removing difficult categories after results are observed. We retain all four project classes in UA-DETRAC-G2 and all ten native scored classes in VisDrone. Because the study contains only three paired seeds and some categories have relatively few instances, per-class contrasts and paired intervals are interpreted descriptively rather than as tests of statistical significance.

3. Methods

3.1 Study design

We designed a matched-control study to distinguish the effect of adding a gated sequence branch from the effect of its Mamba mixer. Three nano-scale YOLO26 detector variants were compared. C0 was the unmodified baseline. C1-W added one single-axis packed bidirectional branch at the first top-down P4 fusion. C1-C8 preserved the C1-W insertion site and branch shell but replaced only the packed

bidirectional Mamba mixer with a near-parameter-matched local Conv1d mixer. For a given dataset and seed, all compatible baseline parameters, optimization settings, augmentations, checkpoint rules, and evaluators were fixed across variants. Fig. 1 summarizes the shared architecture, mixer-only intervention, parameter matching, and three predefined contrasts.

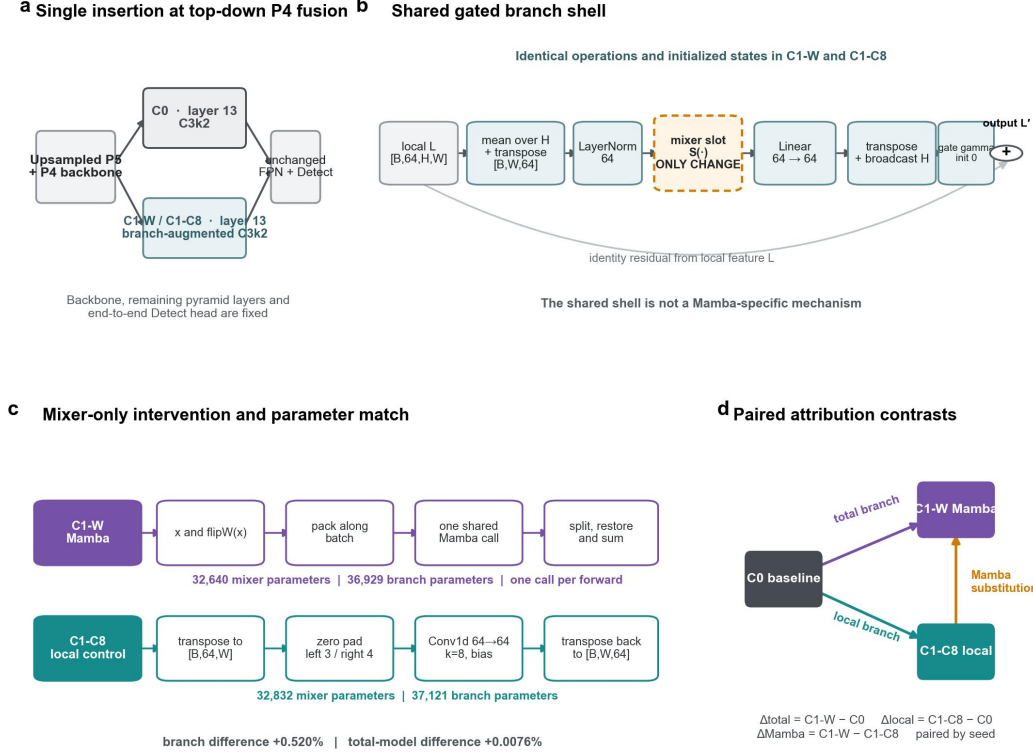


Figure 1 | Matched-control architecture and mixer-only attribution intervention.

The study comprised two independently executed three-variant-by-three-seed experiments. UA-DETRAC-G2 used a sequence-disjoint development split followed by one-time evaluation on the original 40-sequence test partition. VisDrone2019-DET used the official train and validation splits followed by one-time evaluation on test-dev. In both experiments, the nine development-selected best checkpoints, source and data identities, evaluator settings, and analysis rules were frozen before held-out access. Held-out metrics were not used for model selection or further design changes.

3.2 Datasets and partition governance

3.2.1 UA-DETRAC-G2

UA-DETRAC contains 100 traffic videos with vehicle bounding boxes and attributes for detection and tracking evaluation (Wen et al., 2020). We used a project conversion with four labels: car, bus, van, and others. The 60 sequences in the original training pool were divided without sequence overlap into 48 training sequences and 12 development-validation sequences. A deterministic best-improving one-sequence-swap procedure with seed 42 first required every class to be present in validation and then minimized deviations from a target 20% validation share across images and class instances.

The training split contained 6,918 images and 47,738 instances; development validation contained 1,721 images and 11,991 instances (Table M1). The original 40-sequence test partition contained 2,231 images and shared no sequence with either development split. The training runner exposed only the training and development-validation manifests. The test manifest and evaluator were introduced only after all nine checkpoints and confirmatory rules had been frozen.

Table M1 | UA-DETRAC-G2 development data.

Split	Sequences	Images	Car	Bus	Van	others	Total instances
Training	48	6,918	40,256	2,685	4,506	291	47,738
Development validation	12	1,721	10,119	669	1,130	73	11,991

3.2.2 VisDrone2019-DET

VisDrone is a large-scale drone-captured benchmark spanning object detection and tracking tasks (Zhu et al., 2022). We used VisDrone2019-DET with its ten native scored categories: pedestrian, people, bicycle, car, van, truck, tricycle, awning-tricycle, bus, and motor. Official ignored rows, invalid rows, and category 11 others were excluded according to the frozen conversion contract; no category was merged or relabeled.

The official training split contained 6,471 images and 343,204 retained instances after exclusion of 10,345 invalid or ignored rows and one zero-height box. The official validation split contained 548 images and 38,759 retained instances after exclusion of 1,410 invalid or ignored rows. No prepared image had an empty label. Source archives, image headers, annotation fields, geometry, image-label pairing, class counts, and source/prepared image hashes were independently audited before training. Test-dev remained unopened during development.

After the nine formal runs and the converter, evaluator, summarizer, checkpoint hashes, and decision rules had been frozen, the official test-dev archive was converted once under the same ten-class taxonomy. The resulting held-out split contained 1,610 images and 75,102 retained instances. Train/validation artifacts were not modified during test-dev conversion.

3.3 Baseline and branch insertion site

C0 used YOLO26n with its native end-to-end Detect head, `reg_max=1`, and output strides of 8, 16, and 32 pixels (Jocher et al., 2026). An 80-class pretrained checkpoint was transferred wherever tensor names and shapes were compatible; classification-specific tensors were reinitialized for four classes in UA-DETRAC-G2 and ten classes in VisDrone. C0 contained 2,505,360 unfused detector parameters in the four-class implementation.

For C1-W and C1-C8, the ordinary C3k2 block at model layer 13 was replaced with a branch-augmented implementation. This layer follows the first top-down upsampling and concatenation with the P4 backbone feature. The original C3k2 computation, all remaining backbone and feature-pyramid layers, and the Detect head were unchanged. The branch acted only on the final local hidden feature before the block output projection.

3.4 Shared gated width-sequence branch

Let $(L \in \mathbb{R}^{B \times C \times H \times W})$ denote the final local hidden feature of the modified C3k2 block, with $C=64$. The branch averaged (L) over height, transposed the remaining width axis into a token sequence, and applied LayerNorm over channels:

$$T = \text{LN}(\text{Mean}_H(L)^T), \quad T \in \mathbb{R}^{B \times W \times C}$$

For sequence mixer $(S(\cdot))$, square projection $(P: \mathbb{R}^C \rightarrow \mathbb{R}^C)$, and scalar gate γ , the branch output was returned to channel-first form, broadcast over height, and added residually:

$$L' = L + \gamma \text{Broadcast}_H(P(S(T))^T)$$

The scalar gate was initialized exactly to zero. C1-W and C1-C8 therefore reproduced the C0 output at initialization while retaining a nonzero gate gradient. The updated (L') replaced only the final hidden feature in the original C3k2 concatenation; the outer C3k2 output convolution was unchanged.

3.5 Mamba and matched local mixers

3.5.1 Packed bidirectional Mamba (C1-W)

C1-W processed the normalized width sequence in forward and reverse order with shared Mamba parameters. The original and reversed sequences were concatenated along the batch dimension, processed in one Mamba-module call, separated, restored to a common order, and summed:

$$S_M(T) = M(T) + \text{Rev}(M(\text{Rev}(T))).$$

The equation denotes two halves of one packed call rather than independent mixers. The configuration was $d_model=64$, $d_state=16$, $d_conv=4$, and $expand=2$. The mixer contained 32,640 parameters. Including LayerNorm, the 64-by-64 projection, and the gate, the branch contained 36,929 parameters, and the four-class detector contained 2,542,289 parameters.

3.5.2 Kernel-8 Conv1d control (C1-C8)

C1-C8 changed only $(S(\cdot))$. Tokens were transposed to $([B, 64, W])$, padded by three positions on the left and four on the right, and processed by a bias-enabled $\text{Conv1d}(64, 64, \text{kernel_size}=8, \text{stride}=1, \text{groups}=1)$. The result was transposed back to $([B, W, 64])$. No activation, additional normalization, skip connection, dilation, grouping, channel expansion, attention operation, or learnable padding was included in the control mixer.

The Conv1d mixer contained 32,832 parameters, 192 more than Mamba (+0.588%). Its complete branch contained 37,121 parameters (+0.520% relative to C1-W), and its four-class detector contained 2,542,481 parameters. The total detector parameter difference between C1-C8 and C1-W was 0.0076%.

3.6 Matched initialization and correctness gates

Training seeds were 0, 1, and 2. For seed (s) , detector construction used the isolated seed $(260715+s)$. Variants were independently reconstructed from the same frozen YOLO26n checkpoint. All compatible shared tensors were required to have identical names, shapes, dtypes, and values. C1-W and C1-C8 also shared identical LayerNorm, projection, and gate states; only mixer state differed. Construction restored the caller's CPU and CUDA random-number-generator states.

Before formal training, local and CUDA tests verified parser registration, shared-state equality, zero-gate agreement with C0, output shapes, one packed mixer call, finite FP32 and native-AMP forward/backward paths, closed- and open-gate gradients, strict checkpoint reload, and checkpoint reconstruction. Each dataset-specific CUDA gate additionally ran a one-epoch AMP smoke test on 2% of training data and required updates to the gate and mixer plus complete validation. Smoke-test metrics were excluded from scientific analysis.

3.7 Training and checkpoint selection

Every formal run used 100 epochs, input size 640, batch size 16, AdamW with initial learning rate 0.00125, final learning-rate factor 0.01, $\beta_1=0.9$, $\beta_2=0.999$, and weight decay 0.0005. Linear warm-up lasted three epochs with momentum 0.8 and zero warm-up bias learning rate. Automatic mixed precision and deterministic execution were enabled. Eight data-loader workers used the spawn start method. Caching, multi-scale training, rectangular batching, cosine decay, and compilation were disabled.

Frozen augmentations used HSV gains 0.015, 0.7, and 0.4; translation 0.1; scale 0.5; horizontal-flip probability 0.5; and mosaic probability 1.0. Rotation, shear, perspective, vertical flipping, MixUp, CutMix, and copy-paste were zero. Mosaic was disabled for the final ten epochs. No dataset-specific hyperparameter search was performed for VisDrone; retaining the 640-pixel input was part of the frozen replication contract.

Runs were executed sequentially in seed-major order, C0 then C1-W then C1-C8, under an exclusive GPU lock. Checkpoints were retained every ten epochs together with best.pt and last.pt. For each run, the selected checkpoint was the earliest checkpoint attaining the maximum unchanged Ultralytics fitness value on development validation. Pairwise differences and held-out results never entered checkpoint selection. Complete development trajectories and the frozen selected epochs are reported in Supplementary Fig. S1.

3.8 One-time held-out evaluation

For UA-DETRAC-G2, the nine frozen best checkpoints were evaluated once on all 2,231 images from the original 40-sequence test partition. The validator used split=test, input size 640, batch size 16, FP32 weights, device 0, eight workers, no test-time augmentation, no compilation, and no plot or prediction export. It was required to return all four classes.

For VisDrone, the corresponding nine checkpoints were evaluated once on all 1,610 test-dev images in the same seed-major order. Settings were split=test, input size 640, batch size 16, FP32 validation, device 0, eight workers, no augmentation or test-time augmentation, no rectangular batching, no compilation, and no plot or prediction export. The validator was required to return all ten classes.

Campaign start permanently ended the sealed state for each held-out partition. Completed results could not be overwritten. Technical interruption permitted resumption only for missing checkpoint results under unchanged manifests, hashes, settings, and environment identity. Neither held-out campaign authorized checkpoint reselection, threshold tuning, class removal, architecture modification, or additional training.

3.9 Metrics and attribution contrasts

The frozen validator reported precision, recall, mAP at intersection-over-union 0.50 (mAP50), and mAP averaged from intersection-over-union 0.50 to 0.95 in steps of 0.05 (mAP50-95). Overall mAP was

macro-averaged across four UA-DETRAC-G2 classes or ten VisDrone classes. Per-class values were retained as secondary descriptive evidence.

Three seed-paired contrasts were predeclared:

1. total Mamba-branch effect: C1-W minus C0; 2. matched local-branch effect: C1-C8 minus C0; 3. Mamba substitution effect: C1-W minus C1-C8.

For each metric and partition, we retained the three raw differences and summarized their mean, median, sample standard deviation, minimum, maximum, and positive-seed count. We also computed the descriptive paired interval

$$\bar{d} \pm t_{0.975,2} \frac{s_d}{\sqrt{3}}, \quad t_{0.975,2} = 4.303.$$

Because only three seed pairs were available, the interval describes uncertainty and is not used for hypothesis testing. Images or objects were not treated as independent replicates, no multiple-testing claim was made, and statistical significance was not authorized. Results from development and held-out partitions were kept separate before cross-partition interpretation.

3.10 Native-forward efficiency protocol

Efficiency was measured before formal C1-C8 training in two position-balanced matrices: C0 versus C1-C8 and C1-W versus C1-C8. Each matrix contained four repetitions in FP32 and FP16, with order alternated across repetitions and each measurement executed in a fresh process. The primary endpoint was the median of four process-level median latencies for FP16 native-forward execution.

Models were fused and evaluated at batch size 1 using a resident ([1,3,640,640]) CUDA tensor. Fifty warm-up iterations preceded 200 timed iterations. CUDA events and per-iteration synchronization were used. The measurement included only eager-PyTorch native end-to-end model forward and excluded disk input, preprocessing, host-to-device transfer, result construction, rendering, saving, and non-maximum suppression. TF32 and torch.compile were disabled; cuDNN benchmarking was enabled. This four-class measurement is reported as an architecture-level engineering comparison and is not presented as VisDrone end-to-end latency.

3.11 Reproducibility and scope

The evidence records include resolved training arguments, source and data SHA256 identities, environment metadata, construction and train-start fingerprints, CUDA-gate results, checkpoint hashes, process-level timing results, held-out campaign manifests, and immutable start/completion records. Training used Python 3.12.3, PyTorch 2.5.1+cu124, CUDA runtime 12.4, Ultralytics 8.4.95, Mamba-SSM 2.2.6.post3, and causal-conv1d 1.5.4 on NVIDIA GeForce RTX 4090 GPUs. Driver and host identity were recorded and validated within each campaign.

The design supports attribution within one YOLO26n insertion site, one Mamba configuration, one matched local control, two datasets, and three seeds. It does not establish universal superiority or inferiority of Mamba, significance of small paired effects, robustness to other placements or scales, or end-to-end deployment performance.

4. Results

4.1 Construction and evaluation contracts passed

The three detector variants satisfied the predeclared construction and evaluation contracts before formal comparison. C0 retained the official YOLO26n path, C1-W added the zero-initialized gated width-sequence branch with packed bidirectional Mamba mixing, and C1-C8 changed only the mixer inside the same branch to a near-parameter-matched kernel-8 Conv1d. The C1-W and C1-C8 detectors contained 2,542,289 and 2,542,481 parameters, respectively. Shared-state checks, zero-gate output equality, finite FP32 and AMP gradients, strict checkpoint reload, optimizer updates, and one-epoch CUDA smoke tests passed. Every formal run used 100 epochs, seeds 0, 1, and 2, and the earliest maximum under the unchanged Ultralytics fitness rule. The held-out campaigns evaluated exactly the resulting nine best checkpoints without additional training or threshold adjustment.

Table 1 | Cross-partition mAP50-95 and paired attribution contrasts. Model values are mean +/- sample SD across three seeds. Contrast values are paired means, where total Mamba branch is C1-W minus C0, matched local branch is C1-C8 minus C0, and Mamba substitution is C1-W minus C1-C8.

Dataset and partition	C0	C1-W	C1-C8	Total Mamba branch	Matched local branch	Mamba substitution
UA-DETRAC-G2 development	0.632996 +/- 0.019158	0.634052 +/- 0.006917	0.642108 +/- 0.012756	+0.001056	+0.009112	-0.008056
UA-DETRAC-G2 held-out test	0.382510 +/- 0.006178	0.384501 +/- 0.011149	0.373510 +/- 0.012454	+0.001992	-0.008999	+0.010991
VisDrone development	0.169658 +/- 0.001293	0.170148 +/- 0.001479	0.170554 +/- 0.000610	+0.000489	+0.000896	-0.000407
VisDrone test-dev	0.140028 +/- 0.000610	0.137899 +/- 0.002551	0.138442 +/- 0.000978	-0.002129	-0.001585	-0.000544

4.2 UA-DETRAC-G2 development results favored the matched local branch, but seed effects varied

On the UA-DETRAC-G2 development validation set, mean mAP50-95 was 0.632996 +/- 0.019158 for C0, 0.634052 +/- 0.006917 for C1-W, and 0.642108 +/- 0.012756 for C1-C8. The total Mamba-branch effect, C1-W minus C0, averaged +0.001056. The matched local-branch effect, C1-C8 minus C0, averaged +0.009112, with seed differences of +0.039108, -0.023817, and +0.012045. The Mamba substitution effect, C1-W minus C1-C8, averaged -0.008056 and was positive for one of three seeds. Thus, the highest development mean belonged to the matched local control rather than the Mamba variant, but the local-branch difference was heterogeneous across seeds.

The sparsely represented others class produced a larger development contrast but greater uncertainty. C1-C8 minus C0 averaged +0.053162 mAP50-95 for others, whereas C1-W minus C1-C8 averaged -0.014322. These values came from only 73 development instances and three seed pairs. They were therefore retained as exploratory class-wise observations rather than evidence of a stable minority-class improvement.

4.3 The UA-DETRAC-G2 held-out test reversed the development mixer ordering

The one-time UA-DETRAC-G2 held-out campaign evaluated all nine frozen checkpoints on 40 sequences and 2,231 images. Mean held-out mAP50-95 was 0.382510 $\pm$ 0.006178 for C0, 0.384501 $\pm$ 0.011149 for C1-W, and 0.373510 $\pm$ 0.012454 for C1-C8. The total Mamba-branch effect was +0.001992, but only one of three paired seed differences was positive. The matched local-branch effect changed from +0.009112 in development to -0.008999 on held-out sequences. The Mamba substitution effect changed from -0.008056 to +0.010991. The predeclared held-out decision was therefore that the development attribution did not replicate; neither the local-branch direction nor the development Mamba-substitution direction passed the held-out decision rules.

4.4 VisDrone development differences were small

The independent VisDrone replication used 6,471 training images with 343,204 retained instances and 548 development validation images with 38,759 retained instances across ten native classes. Mean development mAP50-95 was 0.169658 $\pm$ 0.001293 for C0, 0.170148 $\pm$ 0.001479 for C1-W, and 0.170554 $\pm$ 0.000610 for C1-C8. The paired total Mamba-branch, matched local-branch, and Mamba substitution effects were +0.000489, +0.000896, and -0.000407, respectively. Although the ordering again placed C1-C8 slightly above C1-W and C0, every mean difference was below 0.001 mAP50-95.

4.5 VisDrone test-dev did not preserve the development branch advantage

After the VisDrone development protocol, nine checkpoints, summary code, and decision rules were frozen, the official test-dev archive was accessed once. Conversion retained 75,102 instances across 1,610 images, and all content and campaign integrity gates passed. Mean test-dev mAP50-95 was 0.140028 $\pm$ 0.000610 for C0, 0.137899 $\pm$ 0.002551 for C1-W, and 0.138442 $\pm$ 0.000978 for C1-C8. C1-C8 minus C0 was negative for all three seeds and averaged -0.001585. C1-W minus C0 averaged -0.002129 and was positive for one seed. C1-W minus C1-C8 averaged -0.000544 and was also positive for one seed. The frozen primary outcome favored the local control over Mamba on VisDrone, but neither branch outperformed C0 and no significance claim was authorized.

4.6 Cross-dataset synthesis identified a generalization boundary rather than a superior mixer

The same qualitative development-to-held-out pattern occurred for the matched local branch in both datasets. C1-C8 exceeded C0 by +0.009112 on UA-DETRAC-G2 development and +0.000896 on VisDrone development, but fell below C0 by -0.008999 on the UA-DETRAC-G2 held-out test and -0.001585 on VisDrone test-dev. The Mamba substitution direction was not universal: it reversed on UA-DETRAC-G2 and remained slightly negative on both VisDrone partitions. Across the four partitions, the C1-W minus C1-C8 effects were -0.008056, +0.010991, -0.000407, and -0.000544. The experiments therefore did not identify a branch or mixer with a reproducible accuracy advantage across development and held-out evaluation. Fig. 2 summarizes the model-level and paired-seed effects across both datasets and partitions; held-out per-class contrasts are reported descriptively in Supplementary Fig. S2.

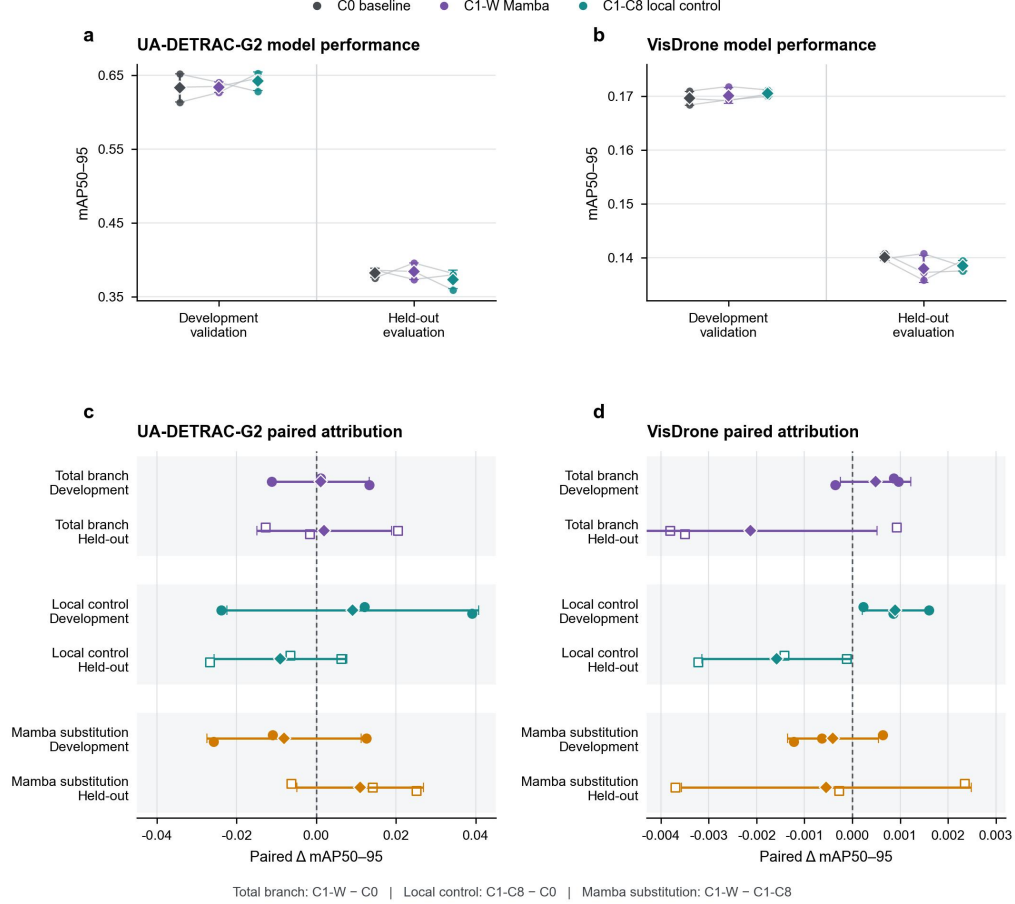


Figure 2 | Branch-level development effects do not generalize consistently across held-out partitions.

4.7 The matched local mixer was computationally cheaper than Mamba under the frozen scope

The position-balanced RTX 4090 benchmark measured synchronized FP16 native-forward latency at batch 1, excluding preprocessing, host-to-device transfer, and non-maximum suppression. Median-of-medians latency was 6.399 ms for C0 and 6.832 ms for C1-C8 in the baseline-control matrix, corresponding to a 6.777% overhead. In the control-Mamba matrix, C1-C8 required 6.828 ms and C1-W required 7.903 ms, making the local control 13.600% faster than the Mamba variant. This result establishes a cost difference under the tested eager-PyTorch scope, not end-to-end deployment latency. The fresh-process measurements and timing scope are shown in Fig. 3.

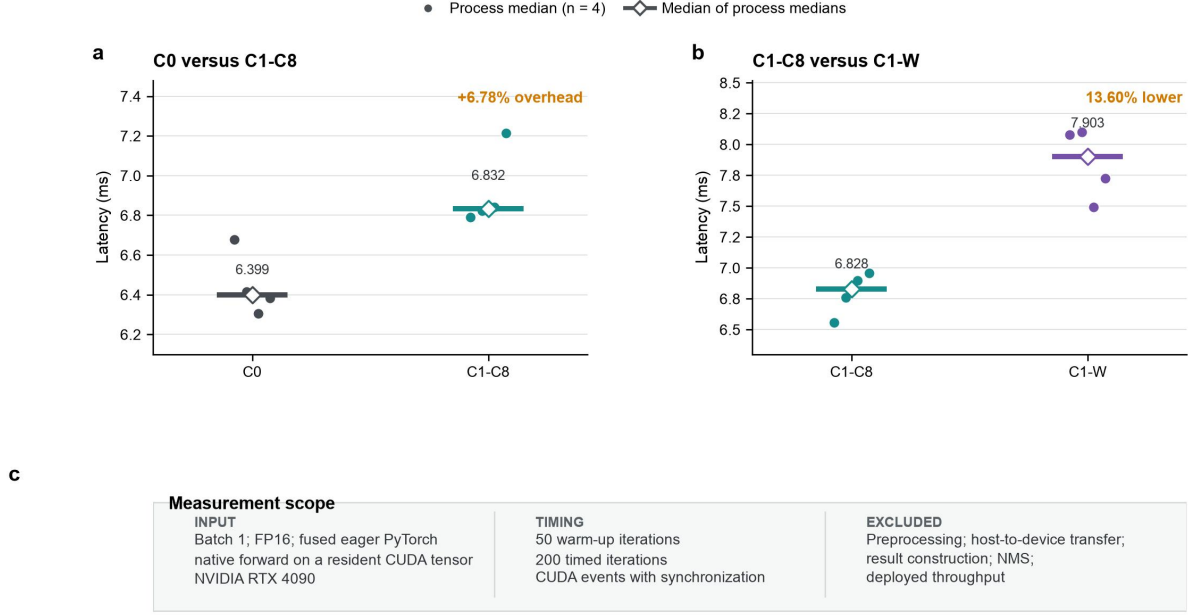


Fig. 3 | Native-forward latency under matched architecture comparisons. a, C0 versus C1-C8 in the position-balanced local-control matrix. b, C1-C8 versus C1-W in the independently measured Mamba-control matrix. Filled dots show the median latency from each fresh process ($n = 4$ per model per comparison); the horizontal line and open diamond show the median of the four process-level medians. C1-C8 is 6.78% slower than C0 in panel a and 13.60% lower in latency than C1-W in panel b. c, Measurement scope: batch 1, FP16, fused eager PyTorch native forward on a resident CUDA tensor, 50 warm-up iterations and 200 timed iterations on an NVIDIA RTX 4090. Preprocessing, host-to-device transfer, result construction and NMS are excluded, so the measurements are native-forward latency rather than end-to-end inference or deployed throughput. The center statistic is the median of process medians; no spread interval or inferential test is displayed. Source data are provided as a Source Data file.

Table 2 | Batch-1 FP16 native-forward latency. Values are median-of-medians over four fresh-process repetitions per model within each position-balanced matrix.

Matrix	First model	Latency (ms)	Second model	Latency (ms)	Relative difference
Baseline-control	C0	6.399	C1-C8	6.832	C1-C8 +6.777% vs C0
Control-Mamba	C1-C8	6.828	C1-W	7.903	C1-C8 -13.600% vs C1-W

5. Discussion

5.1 Matched controls changed the attribution question

The principal finding is not that one additional branch improved YOLO26n, but that the explanation attached to a development-set gain was sensitive to both the comparator and the evaluation partition. On UA-DETRAC-G2 development, the complete Mamba branch remained near C0 while the matched Conv1d branch had the highest mean mAP50-95. A baseline-only comparison could have encouraged the surrounding gated sequence branch to be described as a Mamba benefit. Holding tokenization, normalization, projection, gating, initialization, training, and evaluation fixed changed that interpretation:

the development ordering favored the local control rather than Mamba. Held-out evaluation then showed that even this revised local-branch attribution was not stable.

5.2 Development improvements did not predict held-out effects

The matched local branch exceeded C0 on both development partitions but fell below C0 on both held-out partitions. The reversal was larger on UA-DETRAC-G2 and smaller on VisDrone, yet the direction was shared. This repeated pattern is consistent with a development-specific interaction among the branch, checkpoint selection, and partition composition. It does not identify which of these factors caused the reversal. In particular, the experiments do not show that pooling, LayerNorm, projection, residual gating, local smoothing, or any measured data characteristic was individually responsible. They show only that a small branch-level development advantage was insufficient evidence of held-out improvement.

The Mamba substitution contrast provided a narrower test. It reversed from negative to positive between UA-DETRAC-G2 development and held-out evaluation, whereas it remained negative but close to zero on both VisDrone partitions. Consequently, neither a universal Mamba advantage nor a universal local-mixer advantage is supported. The VisDrone consistency is useful because it independently repeats the absence of a Mamba-specific development advantage, but its effect size is too small and its three-seed sample too limited to support a significance claim.

5.3 The result complements rather than invalidates complete-system Mamba detectors

Representative Mamba-based detectors combine state-space processing with macro-architecture, residual, channel-interaction, or cross-layer-fusion changes (Wang et al., 2025; Chen et al., 2026). Such studies evaluate complete systems, whereas the present experiment asks whether one mixer contributes beyond a fixed surrounding branch. The negative mixer-specific result therefore does not imply that state-space models are ineffective in detection. It shows that a gain from a multi-component Mamba-labelled block cannot be assigned to its selective state-space mixer unless a matched non-Mamba control is included. This distinction is especially important when the complete branch changes capacity, tokenization, normalization, or fusion at the same time.

5.4 Efficiency strengthened the practical case for matched controls, not for a new model

C1-C8 was 13.600% faster than C1-W under the frozen batch-1 FP16 native-forward protocol while retaining nearly the same parameter count and branch shell. This cost difference makes the local mixer a meaningful engineering control rather than a deliberately weak ablation. However, neither candidate produced a held-out accuracy benefit in both datasets. The latency result should therefore be interpreted as evidence that mixer choice affects computational cost even when accuracy differences are unstable, not as a reason to promote C1-C8 as a new detector.

5.5 Class-wise signals require restraint

The UA-DETRAC-G2 development others contrast originally motivated closer examination because its mean change was larger than the aggregate effect. That class contained only 73 development instances and was heterogeneous by construction. The corresponding held-out evidence did not authorize a stable minority-class claim. VisDrone provided ten native categories and a substantially larger evaluation set, but its overall mixer contrasts remained small and per-class differences were not used for test-based architecture selection. These observations support class-wise reporting as a diagnostic, while cautioning against constructing a general minority-class mechanism from one sparse label.

5.6 Limitations and applicability boundary

Four boundaries constrain the interpretation. First, each partition-level contrast contains only three paired seeds, so the intervals are descriptive and no significance claim is authorized. Second, the study covers two traffic-oriented datasets, one YOLO26n insertion site, one packed bidirectional Mamba configuration, and one kernel-8 Conv1d control; it does not represent all state-space detectors or non-Mamba mixers. Third, checkpoints were selected by development validation performance before one-time held-out evaluation. This design protects the test partitions from tuning but does not eliminate selection variance. Fourth, latency was measured on one RTX 4090 software stack under native-forward eager-PyTorch execution, not as end-to-end deployment latency. Fifth, the study estimates within-architecture component effects rather than detector leaderboard performance; it neither retunes YOLO26n against unrelated detectors nor treats absolute accuracy as evidence of state-of-the-art performance.

Within these boundaries, the study suggests a concrete evaluation practice. A detector paper that attributes a gain to Mamba should report at least the unchanged baseline, the complete Mamba branch, and a capacity- and shell-matched non-Mamba mixer under paired seeds. The same frozen checkpoints should then be assessed on an independent partition or dataset without model reselection. This design does not guarantee a positive result, but it distinguishes a mixer effect from a branch effect and exposes whether the inferred direction survives beyond development data.

6. Conclusion

This study isolated packed bidirectional Mamba mixing from the surrounding gated width-sequence branch in YOLO26n using a near-parameter-matched Conv1d control, three paired seeds, two traffic-object datasets, and one-time held-out evaluation. The matched local branch had the highest development mean in both datasets, but its advantage over C0 became negative on both held-out partitions. Mamba likewise showed no reproducible advantage over the local control across the four evaluated partitions, while incurring higher native-forward latency under the tested scope. The evidence supports matched mixer controls and independent evaluation as requirements for credible component attribution. It does not establish statistical significance, a generally superior branch, or a universal conclusion about state-space detectors.

References

- Cawley, G.C., and Talbot, N.L.C. (2010). On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation. *Journal of Machine Learning Research*, 11(70), 2079-2107.
<https://www.jmlr.org/papers/v11/cawley10a.html>
- Chen, L., Wang, H., Xu, M., Li, Z., and Fan, Z. (2026). AKCMamba-YOLO: Selective State Space Models For Real-Time Object Detection. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 4438-4447. https://openaccess.thecvf.com/content/CVPR2026/html/Chen_AKCMamba-YOLO_Selective_State_Space_Models_For_Real-Time_Object_Detection_CVPR_2026_paper.html
- Dwork, C., Feldman, V., Hardt, M., Pitassi, T., Reingold, O., and Roth, A. (2015). The Reusable Holdout: Preserving Validity in Adaptive Data Analysis. *Science*, 349(6248), 636-638.
<https://doi.org/10.1126/science.aaa9375>
- Gu, A., and Dao, T. (2024). Mamba: Linear-Time Sequence Modeling with Selective State Spaces. *First Conference on Language Modeling*. <https://doi.org/10.48550/arXiv.2312.00752>

- Gupta, A., Dollar, P., and Girshick, R. (2019). LVIS: A Dataset for Large Vocabulary Instance Segmentation. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. <https://doi.org/10.1109/CVPR.2019.00550>
- Jocher, G., Qiu, J., Liu, M., Lyu, S., Akyon, F.C., and Kalfaoglu, M.E. (2026). Ultralytics YOLO26: Unified Real-Time End-to-End Vision Models. <https://doi.org/10.48550/arXiv.2606.03748>
- Lei, X., Wu, S., Wu, W., and Jiang, Z. (2025). MambaNeXt-YOLO: A Hybrid State Space Model for Real-time Object Detection. *arXiv:2506.03654*. <https://doi.org/10.48550/arXiv.2506.03654>
- Li, Y., and Xiao, J. (2024). DS MYOLO: A Reliable Object Detector Based on SSMs for Driving Scenarios. *arXiv:2409.01093*. <https://doi.org/10.48550/arXiv.2409.01093>
- Lin, T.Y., Goyal, P., Girshick, R., He, K., and Dollar, P. (2017). Focal Loss for Dense Object Detection. *Proceedings of the IEEE International Conference on Computer Vision*, 2980-2988. <https://doi.org/10.1109/ICCV.2017.324>
- Liu, Y., Tian, Y., Zhao, Y., Yu, H., Xie, L., Wang, Y., Ye, Q., Jiao, J., and Liu, Y. (2024). VMamba: Visual State Space Model. *Advances in Neural Information Processing Systems* 37. <https://doi.org/10.52202/079017-3273>
- Shi, Y., Dong, M., and Xu, C. (2024). Multi-Scale VMamba: Hierarchy in Hierarchy Visual State Space Model. *Advances in Neural Information Processing Systems* 37. <https://doi.org/10.52202/079017-0808>
- Tan, J., Wang, C., Li, B., Li, Q., Ouyang, W., Yin, C., and Yan, J. (2020). Equalization Loss for Long-Tailed Object Recognition. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. <https://doi.org/10.1109/CVPR42600.2020.01168>
- Wang, H., He, Q., Peng, J., Yang, H., Chi, M., and Wang, Y. (2024). Mamba-YOLO-World: Marrying YOLO-World with Mamba for Open-Vocabulary Detection. *arXiv:2409.08513*. <https://doi.org/10.48550/arXiv.2409.08513>
- Wang, Z., Li, C., Xu, H., Zhu, X., and Li, H. (2025). Mamba YOLO: A Simple Baseline for Object Detection with State Space Model. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(8), 8205-8213. <https://doi.org/10.1609/aaai.v39i8.32885>
- Wen, L., Du, D., Cai, Z., Lei, Z., Chang, M.C., Qi, H., Lim, J., Yang, M.H., and Lyu, S. (2020). UA-DETRAC: A new benchmark and protocol for multi-object detection and tracking. *Computer Vision and Image Understanding*, 193, 102907. <https://doi.org/10.1016/j.cviu.2020.102907>
- Zhu, P., Wen, L., Du, D., Bian, X., Fan, H., Hu, Q., and Ling, H. (2022). Detection and Tracking Meet Drones Challenge. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(11), 7380-7399. <https://doi.org/10.1109/TPAMI.2021.3119563>
- Zhu, L., Liao, B., Zhang, Q., Wang, X., Liu, W., and Wang, X. (2024). Vision Mamba: Efficient Visual Representation Learning with Bidirectional State Space Model. *Proceedings of the 41st International Conference on Machine Learning*, 235, 62429-62442. <https://proceedings.mlr.press/v235/zhu24f.html>